UFUG 2104
Written Assignment 1

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March 9, 2026

Tips: Please clilck the link below to get to the targeted answers.

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1 Problem 1

(16 pts) The pdf (probability density function) of X is given by

$$f_X(x) = \begin{cases} -\frac{1}{4}x, & -2 \le x < 0 \\ cx^2, & 0 \le x \le 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of the constant c.

(b) Plot the pdf of X from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e., where they intercept the y-axis and x-axis. (c) Find the expression for the cdf (cumulative distribution function) of X. (d) Find the expected value of X. (e) Find the probability of the event $A = \{1 \le x \le 2\}$. (f) Find the conditional pdf of X given A, $f_{X|A}(x|A)$ and plot it from -3 to 4. If you plot by hand, make sure that your plot shows the main features, i.e. where they intercept the y- axis and x-axis. (g) Find the conditional expected value of X given A, $\mathbb{E}[X|A]$

Solution

(a) By the definition of pdf, we have

$$\int_{-\infty}^{\infty} f_X(x)dx = 1$$

and given the property of the function, we have

$$\int_{-\infty}^{\infty} f_X(x)dx = \int_{-\infty}^{-2} 0dx + \int_{-2}^{0} \left(-\frac{1}{4}x\right) dx + \int_{0}^{3} (cx^2)dx + \int_{4}^{\infty} 0dx = 1$$

Which means

$$(-\frac{1}{8}x^2)\Big|_{-2}^{0} + (\frac{1}{3}cx^3)\Big|_{0}^{3} = \frac{1}{2} + 9c = 1$$

$\Rightarrow \boxed{c = \frac{1}{18}}$

(b) The pdf can be graphed as below:

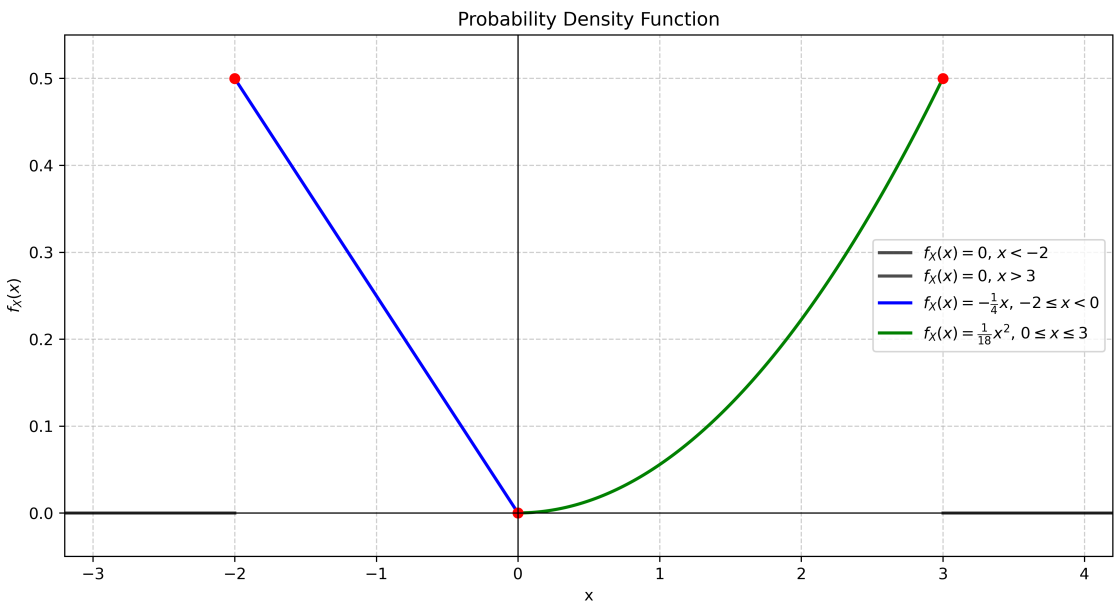

Figure 1

(c) The cdf can be calculated as below:

$$\boxed{F_X(x) = \begin{cases} 0 & x < -2 \\ \frac{1}{2} - \frac{1}{8}x^2 & -2 \le x < 0 \\ \frac{1}{2} + \frac{1}{54}x^3 & 0 \le x \le 3 \\ 1 & 3 < x \end{cases}}$$

(d) the expected value can be cal culated as below: when x < -2, the expected value is 0; when -2 <= x < 0, the expected value is

$$\int_{-2}^{0} -x(\frac{1}{4}x)dx = -\left(\frac{1}{12}x^3\right)\Big|_{x=-2}^{0} = -\frac{2}{3}$$

when $0 \le x \le 3$, the expectation is

$$\int_{0}^{3} x\frac{1}{18}x^2 dx = \left(\frac{1}{72}x^4\right)\Big|_{x=0}^{3} = \frac{9}{8}$$

And the same expectation value of 0m can be applied to the case when $x > 3$. So the total expectation for the expectation of

$$\boxed{\mathbb{E}(X) = -\frac{2}{3} + \frac{9}{8} = \frac{11}{24}}$$

(e) For the event $A = \{1 \le x \le 2\}$, the probability can be calculated in this way:

$$\boxed{\mathbb{P}(A) = \int_{1}^{2} \frac{1}{18}x^2 = \frac{1}{54}x^3\Big|_{x=1}^{2} = \frac{7}{54}}$$

(f) When $x \in [1, 2]$, the conditional pdf can be calculated as below:

$$f_{X|A}(x|A) = \frac{f_X(x)}{\mathbb{P}(A)} = \frac{\frac{1}{18}x^2}{\frac{7}{54}} = \frac{3}{7}x^2, \quad x \in [1, 2]$$

When $x \notin [1, 2]$, the conditional pdf can be calculated as 0. So we have:

$$\boxed{f_{X|A}(x|A) = \begin{cases} \frac{3}{7}x^2 & x \in [1, 2] \\ 0 & \text{otherwise} \end{cases}}$$

(g) we only have to calculate the conditional Expectation when $x \in [1, 2]$:

$$\boxed{\mathbb{E}[X|A] = \int_{1}^{2} \frac{3}{7}x^2 = \frac{9}{7}}$$

2 Problem 2

(16 pts) Suppose we have two machines. The lifetime of Machine 1 in years, X_1, follows an exponential distribution with parameter $\lambda_1 = \frac{1}{9}$. The lifetime of Machine 2 in years, X_2, follows an exponential distribution with parameter $\lambda_2 = \frac{1}{5}$.

(a) Compute the expected values of X_1 and X_2. Which machine lasts longer on average?

(b) Find the conditional pdf of X_2 given that the lifetime of Machine 2 is less than 4 years.

(c) Compute the conditional expected value of X_2 given that the lifetime of Machine 2 is less than 4 years.

(d) Find the probability that the lifetime of Machine 1 exceeds 9 years. Repeat for Machine 2.

(e) Find the probability that Machine 1 will continue to work for at least another 6 years, given that the machine has been working for 9 years already

Solution

(a)

Lemma We have the expectation value:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \lambda e^{-\lambda x} dx = \int_{-\infty}^{0} \lambda e^{-\lambda x} dx + \int_{0}^{\infty} \lambda e^{-\lambda x} dx = 0 + (-e^{-\lambda x})\Big|_0^{\infty} = \frac{1}{\lambda}$$

So the expectation of the Random Variables X_1, X_2 are:

$$\boxed{\mathbb{E}[X_1] = \frac{1}{\lambda_1} = 9, \quad \mathbb{E}[X_2] = \frac{1}{\lambda_2} = 5}$$

(b) By the reality meaning of the exponential distribution of the lifetime of the machine, we have $x \geq 0$ And the probability of $0 < x < 4$, we have

$$\mathbb{P}_X[0 < x < 4] = \int_0^4 \frac{e^{-\frac{x}{5}}}{5} = -e^{-\frac{x}{5}}\Big|_0^4 = 1 - e^{-\frac{4}{5}}$$

And the condotional pdf can be divided to 2 situations: when $0 < x < 4$, we have

$$\mathbb{P}_X[0 < x < x_1] = \int_0^x \frac{e^{-\frac{x_1}{5}}}{5} = 1 - e^{-\frac{x_1}{5}}$$

So the conditional pdf can be written as:

$$\mathbb{P}[X] = \frac{\frac{1}{5}e^{-\frac{x_1}{5}}}{1 - e^{-\frac{4}{5}}}$$

when $x > 4$, and we have:

$$\mathbb{P}[X] = 1$$

So in conclusion, we have

$$\mathbb{P}[X] = \begin{cases} \frac{\frac{1}{5}e^{-\frac{x_1}{5}}}{1-e^{-\frac{4}{5}}} & 0 \leq x < 4 \\ 0 & 4 < x \end{cases}$$

we will have the conditional pdf as below:

(c) We will have the conditional expected value of X_2 given $X_2 < 4$ By definition,

$$\mathbb{E}[X_2|X_2 < 4] = \int_0^{\infty} x f_{X_2|X_2<4}(x) dx = \frac{1}{5(1 - e^{-\frac{4}{5}})} \int_0^4 x e^{-\frac{x}{5}} dx = 25 - 45e^{-\frac{4}{5}}$$

And we can calculate the numerical value

$$\boxed{\mathbb{E}[X_2|X_2 < 4] = \frac{5 - 9e^{-\frac{4}{5}}}{1 - e^{-\frac{4}{5}}}} \approx 1.734\text{years}$$

(d) for the exponential distribution,

$$\mathbb{P}[X > t] = e^{-\lambda t}$$

for machine 1,

$$\mathbb{P}[X_1 > 9] = e^{-1}$$

for machine 2,

$$\mathbb{P}[X_2 > 9] = e^{\frac{9}{5}}$$

So

$$\boxed{\mathbb{P}[X_1 > 9] = e^{-1} \approx 0.3679, \quad \mathbb{P}[X_2 > 9] = e^{-\frac{9}{5}} \approx 0.1653.}$$

(e) We want

$$\mathbb{P}[X_1 > 15 | X_1 > 9] = \frac{\mathbb{P}[X_1 > 15]}{\mathbb{P}[X_1 > 9]}$$

Since $X_1 \propto E\left[\frac{1}{9}\right]$

$$\mathbb{P}(X_1 > 15) = e^{-\frac{5}{3}}, \quad \mathbb{P}(X_1 > 9) = e^{-1}$$

Hence

$$\boxed{\mathbb{P}(X_1 > 15 | X_1 > 9) = e^{-\frac{2}{3}}}$$

3 Problem 3

The joint pdf of X and Y is given by:

$$f_{X,Y}(x, y) = \begin{cases} c(x+y)e^{-y}, & -y \le x \le y, \quad 0 \le y \le \infty \\ 0, & \text{otherwise} \end{cases}$$

For this question, it may be useful to draw the region in the X, Y plane where the pdf is non- zero to help you determine the limits of the integrals.

(a) Find the value of the constant c.

(b) Find the marginal pdfs of X and Y, respectively.

(c) Find the probability that $X > 2$ and $Y > 2$, $P[X > 2 \cap Y > 2]$.

(d) Find the conditional pdf of X given Y, $f_{X|Y}(x|y)$. Plot $f_{X|Y}(x|y)$ for $y = 2$.

(e) Find the conditional expectation of X given $Y = y$, $\mathbb{E}[X|y]$. Plot $\mathbb{E}[X|y]$ for y from 0 to 6.

Solution

(a) We have the integration of the pdf must be 1, so we can integrate the pdf to get the c

$$\int_0^\infty \int_{-y}^{y} c(x+y)e^{-y} dx dy = 1$$

we integrate the x first:

$$\int_{-y}^{y} (x+y) dx = \int_{-y}^{y} x dx + \int_{-y}^{y} y dx = 0 + 2y^2 = 2y^2$$

So the

$$\int_{-y}^{y} = c(x+y)e^{-y} dx = 2cy^2 e^{-y}$$

And substitude it into the original equation,

$$\int_0^\infty 2cy^2 e^{-y} dy = 1$$

By the Gamma function identity, we have

$$\int_0^\infty y^2 e^{-y} dy = 2! = 2$$

So we have

$$\boxed{c = \frac{1}{4}}$$

(b) Consider the marginal of Y,

$$f_Y(y) = \int_{-y}^{y} f_{X,Y}(x,y)dxdy$$

Substituting the $c = \frac{1}{4}$

$$f_Y(y) = \frac{1}{4}e^{-y}\int_{-y} y(x+y)dx = \frac{1}{2}y^2e^{-y}, \quad y \geq 0$$

So the marginal of Y,

$$\boxed{f_Y(y) = \begin{cases} \frac{1}{2}y^2e^{-y} & y \geq 0 \\ 0 & y < 0 \end{cases}}$$

And consider the m,arginal of X, given that $|x| < y$ so

$$f_X(x) = \int_{|x|}^{\infty} f_{X,Y}(x,y)dy$$

Substituting the $c = \frac{1}{4}$

$$f_X(x) = \int_{|x|}^{\infty} \frac{1}{4}(x+y)e^{-y}dy = \frac{1}{4}\left(x\int_{|x|}^{\infty} e^{-y}dy + \int_{|x|}^{\infty} ye^{-y}dy\right)$$

if $x > 0$, then

$$f_X(x) = \frac{1}{4}(2x+1)e^{-x}$$

if $x < 0$, then

$$f_X(x) = \frac{1}{4}\left(xe^x + (1-x)e^x\right) = \frac{1}{4}e^x$$

So the equationwe have

$$\boxed{f_X(x) = \begin{cases} \frac{1}{4}e^x & x < 0 \\ \frac{1}{4}(2x+1)e^x & x \geq 0 \end{cases}}$$

(c)

$$p(X > 2, Y > 2) = \int_2^{\infty}\int_2^{y} \frac{1}{4}(x+y)e^{-y}dxdy$$

integrate the X first,

$$\int_2^{y}(x+y)dx = \left[\frac{x^2}{2} + yx\right]_2^{y} = \frac{3}{2}y^2 - 2y - 2$$

And substitude this formula,

$$\frac{1}{4}\int_2^{\infty}\left(\frac{3}{2}y^2 - 2y - 2e^{-y}dyb\right) = \frac{1}{4}\left(\frac{3}{2}10e^{-2} - 2\cdot 3e^{-2} - 2e^{-2}\right) = \boxed{\frac{7}{4}e^{-2}}$$

(d) given that

$$f_{X,Y}(x,y) = \frac{1}{4}(x+y)e^{-y}, \quad f_Y(y) = \frac{1}{2}e^{-}y \quad y \geq 0$$

So

$$f_{X|Y}(x|y) = \begin{cases} \frac{x+y}{2y^2} & x \leq |y|,\ y > 0 \\ 0 & \text{otherwise} \end{cases}$$

For $y = 2$

$$f_{X|Y}(x|2) = \begin{cases} \frac{x+2}{8} & -2 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

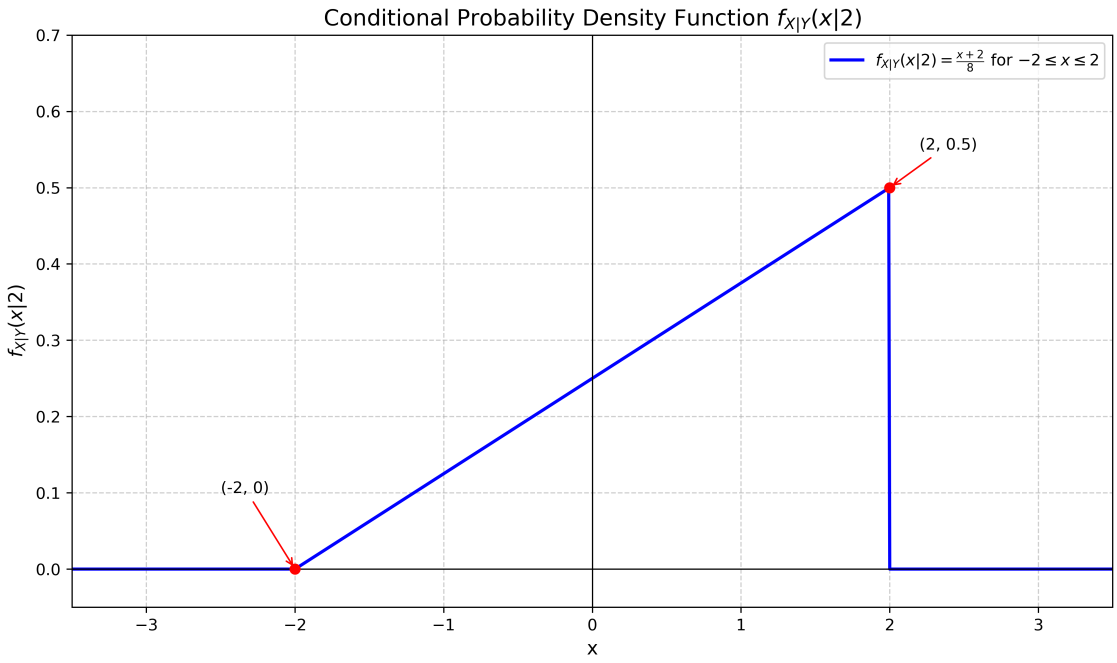

Figure 2

(e) By the conditional density,

$$\mathbb{E}[X|Y = y] = \int_{-y}^{y} x f_{X|Y}(x|y)dx = \int_{-y}^{y} x\frac{x+y}{2y^2}dx = \frac{1}{2y^2}\int_{-y}^{y}(x^2 + xy)dx = \frac{1}{2y^2}\left(\frac{2}{3}y^3\right) = \frac{1}{3}y, \quad y \ge 0$$

$$\boxed{\mathbb{E}[X|Y = y] = \frac{1}{3}y}$$

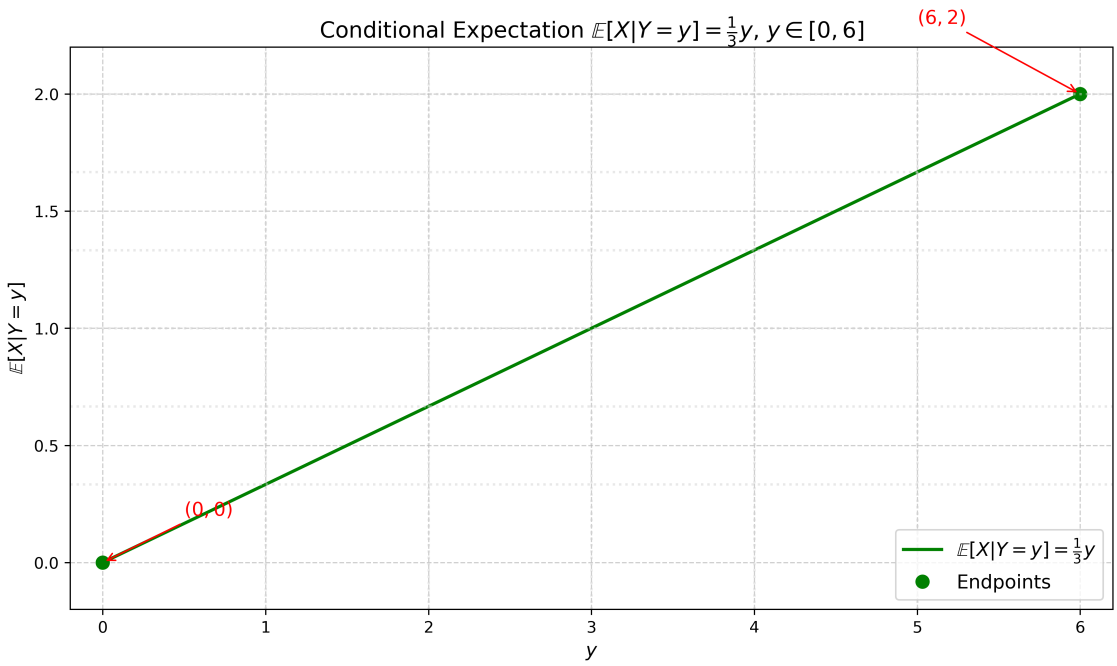

Figure 3

4 Problem 4

(12 pts) Let X and Y be jointly Gaussian random variables with means $\mathbb{E}[X] = 1$ and $E[Y] = -4$, variances $\text{Var}(X) = 4$ and $VarY = 9$. Define a random variable $Z = -2X + Y$.

(a) Assume that X and Y are uncorrelated. Find the pdf of Z, $f_Z(z)$ for $-\infty < z < \infty$.

(b) Assume that X and Y have a correlation coefficient $\rho = -0.3$. Find the pdf of Z, $f_Z(z)$ for $-\infty < z < \infty$.

Solution

(a) Since X and Y are jointly Gaussian, any linear combination of X and Y is also Gaussian. Hence,

$$Z = -2X + Y$$

is Gaussian. Therefore, to determine the distribution of Z, it suffices to compute its mean and variance.

First, by linearity of expectation,

$$\mathbb{E}[Z] = \mathbb{E}[-2X + Y] = -2\mathbb{E}[X] + \mathbb{E}[Y].$$

Substituting the given values,

$$\mathbb{E}[Z] = -2(1) + (-4) = -6.$$

Since X and Y are uncorrelated, we have

$$\text{Cov}\,(()\,X, Y) = 0.$$

Using the variance formula for a linear combination,

$$\text{Var}\,(()\,aX + bY) = a^2\text{Var}\,(()\,X) + b^2\text{Var}\,(()\,Y) + 2ab\,\text{Cov}\,(()\,X, Y),$$

we obtain

$$\text{Var}\,(()\,Z) = \text{Var}\,(()\,-2X + Y) = (-2)^2\text{Var}\,(()\,X) + 1^2\text{Var}\,(()\,Y) + 2(-2)(1)\text{Cov}\,(()\,X, Y).$$

Substituting the given values,

$$\text{Var}\,(()\,Z) = 4 \cdot 4 + 1 \cdot 9 + 2(-2)(1)(0) = 16 + 9 = 25.$$

Thus,

$$Z \sim N(-6, 25).$$

Therefore, the pdf of Z is

$$f_Z(z) = \frac{1}{\sqrt{2\pi(25)}} \exp\left(-\frac{(z-(-6))^2}{2(25)}\right) = \frac{1}{5\sqrt{2\pi}} \exp\left(-\frac{(z+6)^2}{50}\right), \qquad -\infty < z < \infty.$$

(b) Since X and Y are jointly Gaussian, the random variable

$$Z = -2X + Y$$

is also Gaussian. Thus, we again compute its mean and variance.

By linearity of expectation,

$$\mathbb{E}[Z] = -2\mathbb{E}[X] + \mathbb{E}[Y] = -2(1) + (-4) = -6.$$

Next, we use the relation between covariance and correlation:

$$\text{Cov}\,(()\,X, Y) = \rho\,\sigma_X\sigma_Y,$$

where

$$\sigma_X = \sqrt{\text{Var}\,(()\,X)} = \sqrt{4} = 2, \qquad \sigma_Y = \sqrt{\text{Var}\,(()\,Y)} = \sqrt{9} = 3.$$

Since $\rho = -0.3$, we get

$$\text{Cov}\,(()\,X, Y) = (-0.3)(2)(3) = -1.8.$$

Now apply the variance formula for a linear combination:

$$\mathrm{Var}\left(()\, Z\right) = \mathrm{Var}\left(() - 2X + Y\right) = (-2)^2 \mathrm{Var}\left(()\, X\right) + 1^2 \mathrm{Var}\left(()\, Y\right) + 2(-2)(1)\mathrm{Cov}\left(()\, X, Y\right).$$

Substituting the known values,

$$\mathrm{Var}\left(()\, Z\right) = 4 \cdot 4 + 1 \cdot 9 + 2(-2)(1)(-1.8) = 16 + 9 + 7.2 = 32.2.$$

Hence,

$$Z \sim N(-6, 32.2).$$

Therefore, the pdf of Z is

$$f_Z(z) = \frac{1}{\sqrt{2\pi(32.2)}} \exp\left(-\frac{(z-(-6))^2}{2(32.2)}\right) = \frac{1}{\sqrt{64.4\pi}} \exp\left(-\frac{(z+6)^2}{64.4}\right), \qquad -\infty < z < \infty.$$

5 Problem 5